

A brief comparison of adic spaces to other non-archimedean geometries

Eigenvarieties Study Group 2025

1 Introduction

Last week we had a crashcourse on adic spaces, in this talk I want to compare adic spaces to Tate's rigid analytic spaces, Raynaud formal models, and (if time permits) Berkovich spaces.

The basic goal of non-archimedean geometry is to build a theory of analytic geometry over non-archimedean fields such as $\mathbb{Q}_p$. However since $\mathbb{Q}_p$ is totally disconnected as a topological space, the development of this theory is more subtle.

Example 1.1. Let K be an algebraically closed non-archimedean field. Mirroring complex analytic geometry, we would like to define the unit disc

$$\mathbb{D}_K^1 = \{x \in K \mid |x| \leq 1\}$$

and we would like the ring of analytic functions on the unit disc to consist of power series, which converge for $|x| \leq 1$, i.e.

$$\left\{ \sum_{i \geq 0} a_i T^i \mid a_i \rightarrow 0 \text{ for } i \rightarrow \infty \right\}.$$

We already see the difference between this ring and the analogue in complex analytic geometry. We next want to define a sheaf of analytic functions. However, the naive approach of defining the sheaf of functions on an open subset $U \subset \mathbb{D}_K^1$, as functions locally represented as power series, leads to a much larger ring of functions on $\mathbb{D}_K^1$, since $\mathbb{D}_K^1$ is disconnected (for instance, $\{|x| < 1\} \cup \{|x| = 1\}$ is a cover of $\mathbb{D}_K^1$).

In the 1960's, Tate developed the theory of rigid analytic spaces, a theory for analytic geometry over p -adic numbers, locally built out of so called affinoid algebras, with sheaves of analytic functions defined on a Grothendieck topology. Later, more refined theories were developed, in the shape of Berkovich spaces, Raynaud's formal models and Huber's adic spaces.

In the entire note k will be a non-archimedean field.

2 Tate's rigid analytic spaces

Since we want to focus on adic spaces, we will not give a definition of Tate's rigid analytic spaces, for a definition see [Con07]. We will only briefly discuss the idea behind them.

Definition 2.1 (Tate algebra). The n -variable Tate algebra is

$$k\langle T_1, \dots, T_n \rangle := \left\{ \sum_J a_J T^J \mid |a_J| \rightarrow 0 \text{ for } \|J\| \rightarrow \infty \right\}$$

where $J = (j_1, \dots, j_n) \in \mathbb{Z}_{\geq 0}^n$ and $\|J\| = j_1 + \dots + j_n$.

Definition 2.2 (Affinoid algebra). A k -affinoid algebra is a k -algebra A admitting an isomorphism $A \cong k\langle T_1, \dots, T_n \rangle / I$ as k -algebras, for some ideal $I \subset k\langle T_1, \dots, T_n \rangle$.

For an affinoid algebra A we wish to endow the set $\text{MaxSpec}(A)$ with a suitable Grothendieck topology and a sheaf of analytic functions, to obtain affinoid spaces, $\text{Sp}(A)$. We then define rigid analytic spaces as a locally ringed Grothendieck-topologized space locally isomorphic to affinoid spaces. The Grothendieck topology will specify admissible open covers of our set. However, this approach has the drawback of pathological examples, such as nontrivial sheaves that have zero stalks everywhere.

Example 2.3. After reading [Con07], one can look at the following example: Let $\mathbb{D}_K^1 = \text{Sp}(K\langle T \rangle)$ be the unit disc over an algebraically closed non-archimedean field K . For an affinoid subdomain $U \subseteq \mathbb{D}_K^1$, we say that U is *huge* if U contains the complement of finitely many open discs of radii ≤ 1 . We set

$$\mathcal{F}(U) = \begin{cases} \mathbb{Z} & \text{if } U \text{ is huge,} \\ 0 & \text{otherwise.} \end{cases}$$

We can show that $\mathcal{F}$ is a sheaf for the admissible topology. However, $\mathcal{F} \neq 0$ but $\mathcal{F}_x = 0$ for every $x \in D$.

Since we are familiar with adic spaces now, we are curious whether rigid analytic spaces are equivalent to some category of adic spaces. Indeed for affinoid opens $\text{Sp}(A) \mapsto \text{Spa}(A, A^\circ)$ already works, although one has to check that this glues!

Theorem 2.4 ([Hub94]). There exists a fully faithful functor

$$\begin{aligned} \{\text{rigid analytic spaces over } k\} &\longrightarrow \{\text{locally finite type adic spaces over } (k, k^\circ)\} \\ X &\longmapsto X^{\text{ad}} \end{aligned}$$

sending an affinoid open $\text{Sp}(A) \mapsto \text{Spa}(A, A^\circ)$.

Note that this will not be an equivalence of categories, since open subspaces of X^{ad} need not arise from admissible open subsets of X . However, we will from now on view *rigid analytic spaces as locally finite type adic spaces over (k, k°)* , and we also got rid of the pathological Example 2.3.

3 Raynaud's formal models

Raynaud's approach to non-archimedean geometry views rigid-analytic varieties as generic fibers of formal schemes. Rather than following Raynaud's original constructions, we will use the language of adic spaces here.

Formal schemes can be viewed as *ind-schemes*, for instance we can think of a formal scheme over $\mathbb{Z}_p$ as a collection of schemes $(X_n)_{n \geq 1}$ over $\mathbb{Z}/p^n$, together with isomorphisms $X_n \cong X_{n+1} \times_{\mathrm{Spec}(\mathbb{Z}/p^{n+1})} \mathrm{Spec}(\mathbb{Z}/p^n)$. Let us regard the adic unit disc over $\mathbb{Q}_p$, corresponding to the Tate algebra

$$\mathbb{Q}_p\langle T \rangle = (\varprojlim \mathbb{Z}_p[T]/(p^n))\left[\frac{1}{p}\right]$$

Setting $X_n := \mathrm{Spec} \mathbb{Z}_p[T]/(p^n)$, we have a collection of schemes, i.e. a formal scheme over $\mathbb{Z}_p$. More accurately, the corresponding formal scheme should be the colimit in the small category of locally spaces $\mathfrak{X} = \varinjlim X_n$. To get to $\mathbb{Q}_p\langle T \rangle$ we would like to invert p , which intuitively means taking the generic fiber.

First, we want to view formal schemes as adic spaces. For a noetherian ring A , equipped with the I -adic topology, the formal scheme $\mathrm{Spf}(A)$ is then given by the collection $X_n = A/I^n$. To formalize the above construction, we again naively send $\mathrm{Spf}(A) \mapsto \mathrm{Spa}(A, A^\circ)$. Gluing this construction gives us the next theorem.

Theorem 3.1 ([Hub94]). There is a fully faithful functor

$$\begin{aligned} \{\text{locally noetherian formal schemes}\} &\longrightarrow \{\text{adic spaces}\} \\ \mathfrak{X} &\longmapsto \mathfrak{X}^{\mathrm{ad}} \end{aligned}$$

sending $\mathrm{Spf}(A)$ to $\mathrm{Spa}(A, A^\circ)$. We will identify formal schemes with their respective adic schemes.

Next we want to interpret what "taking the generic fiber" means. We need a preliminary definition. We will denote by $\hat{A}$ the I -adic completion of A , if A is endowed with the I -adic topology.

Definition 3.2. A morphism $f : \mathfrak{X} \rightarrow \mathfrak{Y}$ of locally noetherian formal schemes is of *formally finite type* if it is locally on $\mathfrak{X}$ and $\mathfrak{Y}$, given by a morphism of affinoids

$$\mathrm{Spa}(B, B) \longrightarrow \mathrm{Spa}(A, A)$$

induced by a ring homomorphism $A \rightarrow B$ such that there is a quotient map of $\hat{A}$ -algebras:

$$A[[T_1, \dots, T_n]]\langle X_1, \dots, X_m \rangle \twoheadrightarrow \hat{B}.$$

Definition 3.3. Let (k, k°) be an analytic field with pseudouniformizer $\varpi \in k^\circ$. Let $\mathfrak{X}$ be a formal scheme of formally finite type over k° considered as an adic space over $\mathrm{Spa}(k^\circ, k^\circ)$. The generic fiber of $\mathfrak{X}$ is defined as the open subspace

$$\mathfrak{X}_\eta = \{x \in \mathfrak{X} \mid |\varpi(x)| \neq 0\}.$$

If $\mathfrak{X}$ is a (classical) formal scheme we write $\mathfrak{X}_\eta$ for $(\mathfrak{X}^{ad})_\eta$.

Note that $\mathrm{Spa}(k^\circ, k^\circ)$ has two points. The generic point is open and cut out precisely by the condition $|\varpi| \neq 0$. Hence, we see that $\mathfrak{X}_\eta$ is nothing but the preimage in $\mathfrak{X}$ of the generic point of $\mathrm{Spa}(k^\circ, k^\circ)$.

Example 3.4. The generic fiber of $\mathfrak{X} = \mathrm{Spa}(k^\circ\langle T \rangle, k^\circ\langle T \rangle)$ is the rational open subset $R(\frac{\varpi}{\varpi}) = \{x \in \mathfrak{X} \mid |\varpi(x)| \neq 0\}$ which is precisely the unit disc $\mathbb{D}_k^1$.

Example 3.5. The generic fiber of $\mathfrak{X} = \mathrm{Spa}(\mathbb{Z}_p[[T]], \mathbb{Z}_p[[T]])$ is

$$\mathfrak{X}_\eta = \bigcup_{n=1}^{\infty} R\left(\frac{T^n}{p}\right) = \mathring{\mathbb{D}}_{\mathbb{Q}_p}(0, 1).$$

i.e. the open unit disc over $\mathbb{Q}_p$.

The next question is whether, for a rigid analytic variety X over (k, k°) , there always exists a formal scheme $\mathfrak{X}$ such that $\mathfrak{X}_\eta = X$. We call such a formal scheme a *formal model*. It turns out that for qcqs rigid analytic varieties over k , a formal model always exists, and is unique up to forming an admissible blowup. We will not go into more detail here.

Theorem 3.6 (Raynaud). There is an equivalence of categories

$$\begin{array}{c} \{\text{qc } \varpi\text{-torsion free formal } k^\circ\text{-schemes} \\ \text{of tft localized by admissible formal blowups}\} \end{array} \xrightarrow{\sim} \{\text{qcqs rigid } k\text{-spaces}\}$$

$$\mathfrak{X} \longmapsto \mathfrak{X}_\eta^{ad}$$

4 Analytification and GAGA

We take a brief excursion to talk about analytification. One of the original goals of rigid analytic geometry is to provide a target category for an analogue of the analytification functor:

$$\begin{array}{c} \{\text{finite type schemes over } \mathbb{C}\} \longrightarrow \{\text{complex analytic spaces}\} \\ X \longmapsto X^{\text{an}} := X(\mathbb{C}) \end{array}$$

for schemes over non-archimedean local fields.

For an adic space X , let $\underline{X} = (X, \mathcal{O}_X)$ denote the underlying ringed space.

Proposition 4.1 ([Eug24], 10.1). Let $Y \rightarrow X$ be a morphism of schemes that is locally of finite type, Z a stable adic space, and $\underline{Z} \rightarrow X$ a morphism of locally ringed spaces. Then the fiber product $Y \times_X Z$ exists, is a stable adic space, and the projection

$$Y \times_X Z \rightarrow Z$$

is locally of finite type.

Remark 4.2. By fiber product we mean an adic space together with a morphisms $Y \times_X Z \rightarrow Z$, $\underline{Y \times_X Z} \rightarrow Y$ whose underlying locally ringed space $\underline{Y \times_X Z}$ fits into a diagram and satisfies the following universal property for all adic spaces S

$$\begin{array}{ccccc} S & & & & \\ & \searrow \exists! \underline{g} & & \searrow & \\ & \underline{Y \times_X Z} & \longrightarrow & Y & \\ & \downarrow & & \downarrow & \\ & \underline{Z} & \longrightarrow & X & \end{array}$$

$\underline{f}$ (curved arrow from S to $\underline{Z}$)

Proof Sketch. We may assume X and Y are affine, and Z is affinoid:

$$\begin{aligned} X &= \operatorname{Spec} A & Y &= \operatorname{Spec} A[T_1, \dots, T_n]/I \\ Z &= \operatorname{Spa}(B, B^+). \end{aligned}$$

We restrict to the case where B is a Tate ring with pseudouniformizer ϖ . For $k \in \mathbb{N}$ consider Tate rings

$$B\langle \varpi^k T_1, \dots, \varpi^k T_n \rangle, \quad B^+\langle \varpi^k T_1, \dots, \varpi^k T_n \rangle$$

with compatible homomorphisms

$$B\langle \varpi^m T_1, \dots, \varpi^m T_n \rangle \longrightarrow B\langle \varpi^k T_1, \dots, \varpi^k T_n \rangle$$

for $m \leq k$ and

$$B[T_1, \dots, T_n] \longrightarrow B\langle \varpi^k T_1, \dots, \varpi^k T_n \rangle$$

so we can view $I \subset B\langle \varpi^k T_1, \dots, \varpi^k T_n \rangle$. Define

$$B_k := B\langle \varpi^k T_1, \dots, \varpi^k T_n \rangle / IB\langle \varpi^k T_1, \dots, \varpi^k T_n \rangle$$

and B_k^+ as the integral closure of $B^+\langle\varpi^k T_1, \dots, \varpi^k T_n\rangle$ in B_k . The embeddings induce open immersions $\mathrm{Spa}(B_m, B_m^+) \hookrightarrow \mathrm{Spa}(B_k, B_k^+)$ which we can glue along the embeddings to

$$Y \times_X Z = \bigcup_k \mathrm{Spa}(B_k, B_k^+).$$

To show this indeed fulfills the universal property, we need to show for any sheafy Huber pair (C, C^+) with the outer diagram

$$\begin{array}{ccccc} & & C & & \\ & \swarrow \psi & \uparrow & \nwarrow & \\ & B_k & \longleftarrow & A[T_1, \dots, T_n]/I & \\ & \uparrow & & \uparrow & \\ B & \longleftarrow & A & & \end{array}$$

φ (curved arrow from B to C)

that the dashed arrow exists. Let c_i be the images of T_i in C . For big enough k , the elements $\varpi^k c_i$ are all contained in C^+ as C^+ is open and ϖ is topologically nilpotent. Thus, we obtain a well-defined homomorphism

$$\mathrm{Spa}(B_k, B_k^+) \longrightarrow \mathrm{Spa}(C, C^+)$$

mapping $\varpi^k T_i$ to $\varpi^k c_i$, and it is clear from the construction that the corresponding diagram commutes. $\square$

Definition 4.3. Let (k, k^+) be an affinoid field. The analytification of a variety X/k is defined as

$$X^{\mathrm{an}} = X \times_{\mathrm{Spec} k} \mathrm{Spa}(k, k^+).$$

Example 4.4 (The affine line). We start with $\mathbf{A}^1 = \mathrm{Spec}(k[T])$. If we follow the recipe above, we are gluing $\mathrm{Spa}(k\langle\varpi^k T\rangle, k^\circ\langle\varpi^k T\rangle)$ together, i.e. closed unit disks. We therefore obtain the adic affine line.

Similarly, the analytification of $\mathbb{P}_k^1$ is the adic projective line.

4.1 Formal Schemes and analytification

Recall the equivalence of categories sending a formal scheme (identified with the associated adic space $\mathfrak{X}^{\mathrm{ad}}$) to its generic fiber. So starting from a scheme X of finite type over a non-archimedean local field k we can:

1. Take some model $\mathcal{X}$ over k° , and then complete, to obtain a formal scheme $\mathfrak{X}$.
2. Identify with the adic space $\mathfrak{X}^{\mathrm{ad}}$ and take the generic fiber $\mathfrak{X}_\eta^{\mathrm{ad}}$

so this construction also gives us a rigid analytic k -space.

So given a scheme X of finite type over k we have to associated rigid analytic spaces X^{an} and $\mathfrak{X}_\eta^{\text{ad}}$. When do they agree?

Example 4.5 (The affine line). Choose $X = \mathbf{A}_k^1$.

1. We saw that X^{an} is the adic affine line $\mathbf{A}_{(k, k^\circ)}^1$, obtained by gluing closed disks $\text{Spa}(k\langle\varpi^k T\rangle, k^\circ\langle\varpi^k T\rangle)$.
2. A formal model is given by $\text{Spec}(k^\circ[T])$, with completion the formal scheme $\text{Spf}(k^\circ\langle T\rangle)$ which has generic fiber $\text{Spa}(k\langle T\rangle, k^\circ\langle T\rangle)$, the closed unit disk.

So these two disagree!

Theorem 4.6 ([Con07], 3.3.9). For X a separated finite type k -scheme, there is a functorial quasi-compact open immersion of rigid spaces $i_X : \mathfrak{X}_\eta \rightarrow X^{\text{an}}$ that is compatible with fiber products. It is an isomorphism when X is k -proper.

Theorem 4.7 (GAGA, [Köp74]). Let S be a proper algebraic variety over K , with analytification S^{an} . Given a coherent sheaf $\mathcal{F}$ on S , there is a functorial analytification $\mathcal{F}^{\text{an}}$, which is coherent sheaf. Then one has the following:

1. The functor $\mathcal{F} \mapsto \mathcal{F}^{\text{an}}$ is an equivalence of the abelian categories of coherent sheaves on S and on S^{an} , respectively.
2. The two δ -functors $\mathcal{F} \mapsto H^i(S, \mathcal{F})$ and $\mathcal{F} \mapsto H^i(S^{\text{an}}, \mathcal{F}^{\text{an}})$ from coherent sheaves on S to K -vector spaces are isomorphic.

5 Berkovich spaces

In the interest of time, this will be rather short. Berkovich spaces are formed locally from an affinoid algebra by considering

$$\mathcal{M}(A) = \{\text{multiplicative seminorms } v : A \rightarrow \mathbb{R}_{\geq 0}\}$$

endowed with the weakest topology such that $x \mapsto x(f)$ is continuous for all $f \in A$. An adic space is a locally spectral space, and every point has a unique generization. Mapping each point to its unique generization allows us to obtain a Berkovich space from an adic space. Berkovich spaces are especially interesting since for example the affine line is Hausdorff and path-connected. Berkovich spaces are especially useful for studying the uniformisation of curves.

References

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